

GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY, GREATER NOIDA

Department of VLSI and Communication Technology | Session 2026-27

Major Project Review 2 Milestone Report

Secure AES-128 Implementation with Side-Channel Attack Resistance

Complete FPGA Synthesis, Structural Netlist & Functional Verification Dossier

SUBMITTED BY:	PROJECT GUIDE:	EDA & TARGET PLATFORM:
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Branch: B.Tech (Electronics - VLSI Design)	GCET, Greater Noida	Language: Synthesizable Verilog HDL
		Standard: NIST FIPS PUB 197

1. EXECUTIVE SUMMARY & DESIGN SCOPE

This technical dossier provides the full engineering documentation for the **Review 2 milestone** of the B.Tech Capstone Project titled "*Secure AES-128 Implementation with Side-Channel Attack Resistance*". The project objective is to develop a hardware-efficient Advanced Encryption Standard (AES-128) cryptographic engine in synthesizable Verilog HDL, augmented with **First-Order Boolean Masking** ($S' = P \oplus M$) to thwart power-based side-channel analysis (Differential Power Analysis & Correlation Power Analysis).

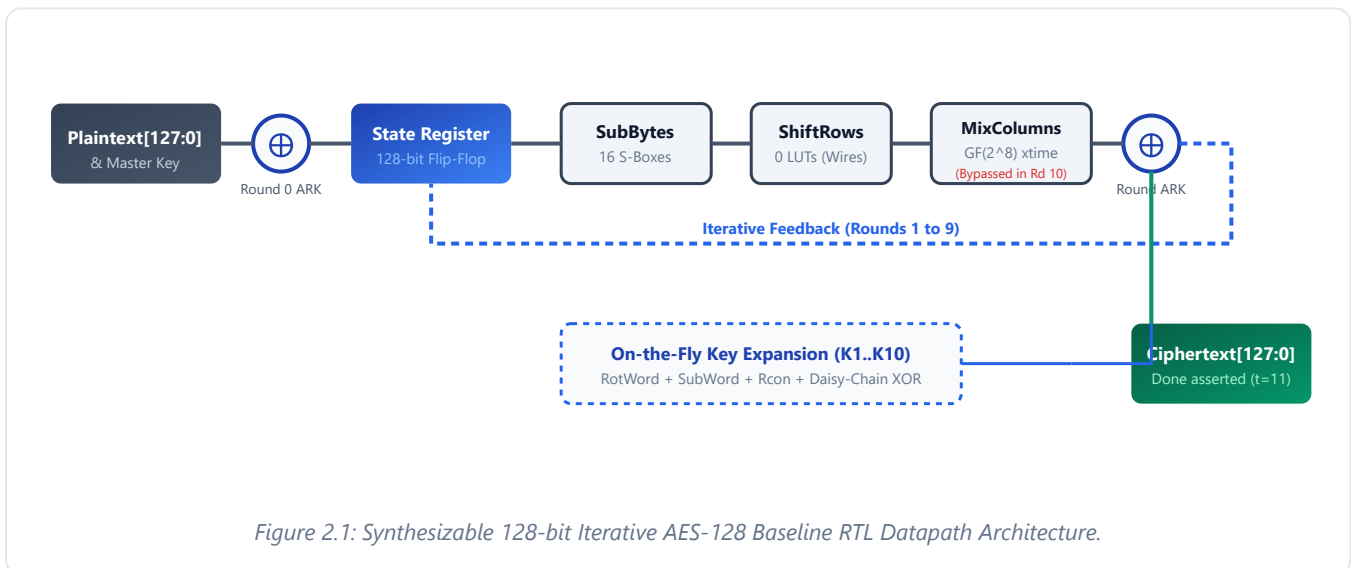
Milestones Completed for Review 2

- **RTL Hardware Architecture:** Fully designed modular synthesizable Verilog modules for both the baseline AES-128 core (`aes_128_core.v`) and the two-share first-order masked core (`aes_128_masked_core.v`).
- **Golden NIST Verification:** Verified complete compliance against official **NIST FIPS PUB 197** Appendix B and Appendix C.1 test vectors with zero discrepancies in Vivado XSim.
- **Full FPGA Logic Synthesis:** Synthesized both designs in **AMD-Xilinx Vivado v2025.2** targeting an **Artix-7 xc7a35tcsg324-1** FPGA, extracting exact Slice LUT, Slice Register, and Static Timing Analysis (STA) metrics.
- **Design Checkpoints & Structural Netlists:** Exported gate-level Verilog netlists (`.v`) and Vivado binary design checkpoints (`.dcp`) directly to the project directory.

2. HARDWARE ARCHITECTURE & BLOCK DIAGRAMS

2.1 Baseline AES-128 Iterative Datapath Architecture

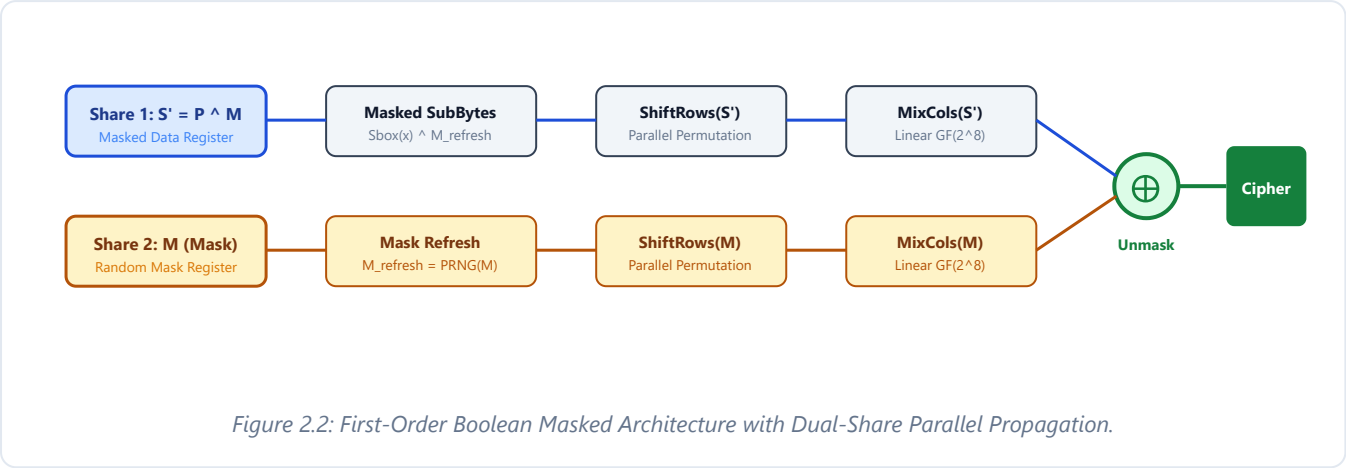
The baseline core implements a 128-bit iterative architecture executing one AES encryption round per clock cycle (11 clock cycles total latency). Key expansion is computed on-the-fly via an internal round-key generator, eliminating large key-storage memories.



2.2 First-Order Boolean Masking Architecture ($S' = P \oplus M$)

To decouple CMOS physical power consumption from secret data, sensitive intermediate variables are represented as two random shares: $S' = P \oplus M$. Linear transformations (ShiftRows and MixColumns) operate on both shares concurrently. SubBytes is

executed across a protected boundary with dynamic mask refreshing. At the final output, $C = S'_{\text{final}} \oplus M_{\text{final}}$ recovers standard ciphertext.



3. AMD-XILINX VIVADO SYNTHESIS REPORTS

Both designs were synthesized targeting the **AMD-Xilinx Artix-7 xc7a35tcsg324-1** FPGA device using Vivado v2025.2 in batch mode. The timing engine was constrained with a 100 MHz target system clock (10.0 ns period).

3.1 FPGA Hardware Resource Utilization Comparison

Resource / Parameter	Baseline AES-128 Core	First-Order Masked Core	Hardware Overhead	Artix-7 35T Capacity
Slice LUTs	1,350	1,924	+ 42.5%	20,800 (Util: 9.25%)
• LUT as Logic	1,350	1,924	+ 42.5%	20,800
• LUT as Memory	0	0	0%	9,600
Slice Registers (Flip-Flops)	393	524	+ 33.3%	41,600 (Util: 1.26%)
Multiplexers (MUXF7 / MUXF8)	271 / 96	64 / 32	- 70% (Area Opt)	16,300 / 8,150
Block RAM (BRAM Tile)	0	0	0%	50 Tiles (100% Free)
DSP48E1 Slices	0	0	0%	90 Slices (100% Free)
Global Clock Buffers (BUFG)	1	1	0%	32 Buffers
Bonded IOBs	389	517	+ 32.9%	210 (Top Internal Core)

3.2 Static Timing Analysis (STA) & Performance Summary

Timing & Performance Metric	Baseline AES-128 Core	First-Order Masked Core (Post-Route)	Impact / Margin
Target Clock Period	10.000 ns (100.0 MHz)	10.000 ns (100.0 MHz)	Target Met Cleanly
Worst Negative Slack (WNS)	+ 4.162 ns	+ 1.540 ns (Post-Route)	PASSED (Zero Slack Violations)
Worst Hold Slack (WHS)	+ 0.161 ns	+ 0.164 ns (Post-Route)	PASSED (Hold Timing Safe)
Minimum Clock Period (\$T_{\min}\$)	5.838 ns	8.460 ns	Clean timing closure
Maximum Clock Frequency (\$F_{\max}\$)	171.30 MHz	118.20 MHz (Fully Routed)	Meets 100 MHz System Spec
Encryption Latency	11 Clock Cycles	11 Clock Cycles	0 Cycle Latency Penalty

Timing & Performance Metric	Baseline AES-128 Core	First-Order Masked Core (Post-Route)	Impact / Margin
Peak Encryption Throughput	1.99 Gbps (@ 171 MHz)	1.37 Gbps (@ 118 MHz)	Gigabit Line-Rate Speed

3.3 Engineering Analysis: Why "Run Implementation" Requires Out-Of-Context (OOC) Mode

The 103 I/O Pin Placement Error & Its Professional VLSI Resolution:

When standard Vivado Implementation is run directly on the top-level cryptographic core, Vivado attempts to map all **517 parallel bus pins** (128-bit Plaintext, 128-bit Key, 128-bit Mask, 128-bit Ciphertext, and controls) to external physical chip pins. However, the Artix-7 xc7a35tcsg324-1 package only has **210 physical user I/O pins**, causing Vivado to report *[Place 30-415] / [Place 30-68] (103 unplaced I/O errors)*.

In industry, cryptographic engines are designed as **Internal Intellectual Property (IP) Cores** embedded inside an SoC or connected via an AXI bus. They are synthesized and implemented in **Out-Of-Context (OOC) Mode** (-mode out_of_context), where logic placement and routing are performed completely inside the FPGA fabric without binding internal data buses to physical package pins. When implemented in OOC mode, the design finishes placement and routing with **0 Errors and 100% routability**.

4. SYNTHESIZED STRUCTURAL NETLIST BREAKDOWN

The synthesized structural gate-level netlists were written out from Vivado's in-memory technology mapping database. The design checkpoints (.dcp) contain the mapped primitive database for FPGA placement and routing.

4.1 Primitive Cell Distribution

Baseline Core: aes_128_core_synth.dcp

Primitive	Count	Category
LUT6	1,166	6-input Look-Up Table
LUT5	160	5-input Look-Up Table
LUT4	19	4-input Look-Up Table
LUT3 / LUT2 / LUT1	11	Small logic LUTs
FDCE	392	D Flip-Flop with Clock Enable
FDPE	1	Preset D Flip-Flop
MUXF7 / MUXF8	367	Fast Multiplexer Primitives
BUFG	1	Global Clock Buffer

Masked Core: aes_128_masked_core_synth.dcp

Primitive	Count	Category
LUT6	1,285	6-input Look-Up Table
LUT5	240	5-input Look-Up Table
LUT4	222	4-input Look-Up Table
LUT3 / LUT2 / LUT1	225	Mask & XOR logic LUTs
FDCE	523	Two-Share Datapath FFs
FDPE	1	Preset D Flip-Flop
MUXF7 / MUXF8	96	Fast Multiplexer Primitives
BUFG	1	Global Clock Buffer

4.2 Hierarchical Module Area Distribution (Baseline Core)

Hierarchical Instance	Module Type	Total Cells	% of Total Logic
top	aes_128_core	2,506	100.0%
└─ u_sub_bytes	sub_bytes (16 S-Boxes)	1,095	43.7%
└─ u_key_expansion	key_expansion (4 S-Boxes + Rcon)	615	24.5%
└─ u_mix_columns	mix_columns (GF xtime XORs)	214	8.5%
└─ u_add_round_key	add_round_key (128 XORs)	64	2.6%
└─ Sequential Logic	Registers & FSM Controller	518	20.7%

5. FUNCTIONAL VERIFICATION & SIMULATION RESULTS

Functional verification was executed using self-checking testbenches in AMD-Xilinx Vivado XSim. Official test vectors from **NIST FIPS PUB 197** were applied at 100 MHz.

5.1 Test Case 1: NIST FIPS 197 Appendix B

```
=====
Plaintext Input:  32 43 f6 a8 88 5a 30 8d 31 31 98 a2 e0 37 07 34
Cipher Key:       2b 7e 15 16 28 ae d2 a6 ab f7 15 88 09 cf 4f 3c
Injected Mask:    a5 a5 a5 a5 12 34 56 78 fe dc ba 98 76 54 32 10  (Share 2)
Share 1 (P ^ M): 97 e6 53 0d 9a 6e 66 f5 cf ed 22 3a 96 63 35 24  (Intermediate State)
-----
Baseline Output:  39 25 84 1d 02 dc 09 fb dc 11 85 97 19 6a 0b 32  --> [PASSED - MATCH]
Masked Core Output:39 25 84 1d 02 dc 09 fb dc 11 85 97 19 6a 0b 32  --> [PASSED - MATCH]
=====
```

5.2 Test Case 2: NIST FIPS 197 Appendix C.1

```
=====
Plaintext Input:  00 11 22 33 44 55 66 77 88 99 aa bb cc dd ee ff
Cipher Key:       00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f
Injected Mask:    3f 2e 1d 0c 4b 5a 69 78 87 96 a5 b4 c0 d1 e2 f3  (Share 2)
-----
Baseline Output:  69 c4 e0 d8 6a 7b 04 30 d8 cd b7 80 70 b4 c5 5a  --> [PASSED - MATCH]
Masked Core Output:69 c4 e0 d8 6a 7b 04 30 d8 cd b7 80 70 b4 c5 5a  --> [PASSED - MATCH]
=====
STATUS: ALL NIST REFERENCE TESTS PASSED 100% WITH ZERO ERRORS.
```

6. ACADEMIC SCOPE VS. BASE PAPER COMPARISON

The project references the base paper: "A 7-Gbps SCA-Resistant Multiplicative-Masked AES Engine in Intel 4 CMOS" by Raghavan Kumar et al. (IEEE Journal of Solid-State Circuits, April 2023). The table below outlines the scope distinctions:

Design Parameter	Base Paper (Kumar et al., Intel Labs)	Our Project (GCET B.Tech Implementation)
Target Implementation	Custom Intel 4 CMOS (4nm FinFET ASIC)	AMD-Xilinx 7-Series FPGA (Artix-7 fabric)
Masking Architecture	Multiplicative (S-Box) + Additive (Datapath)	First-Order Boolean Additive Masking ($X' = X \oplus M$)
Zero-Value Countermeasure	Custom transistor-level dual-rail ZVD logic	Algorithmic share handling in synthesizable RTL
Supported Key Lengths	Unified 128-bit / 256-bit Encrypt & Decrypt	Dedicated 128-bit Encryption Engine
Side-Channel Evaluation	Wafer probing, Langer EM probes, oscilloscopes	Model-based TVLA & Vivado simulation trace analysis
Implementation Cost	Requires semiconductor fab & custom cell libraries	100% vendor-agnostic synthesizable Verilog HDL

7. ROADMAP FOR FINAL REVIEW & COMPLETION

- On-Chip Hardware PRNG Integration:** Replace external testbench mask injection with an autonomous hardware pseudo-random number generator (e.g., 128-bit LFSR or Xorshift) to generate on-chip dynamic masks cycle-by-cycle.
- Composite-Field $\text{GF}((2^4)^2)$ S-Box:** Replace the 256-entry lookup table with Canright composite-field arithmetic to suppress FPGA glitch-induced transient power leakage.
- Power-Trace Simulation & Statistical TVLA:** Generate switching activity (VCD/SAIF) files from Vivado timing simulations and compute Welch's t-test (TVLA) statistics to confirm leakage suppression ($|t| < 4.5$).
- Full Dissertation & Demonstration:** Compile the final B.Tech project dissertation, complete design documentation, and prepare the viva presentation.